

Last name _____

First name _____

LARSON—MATH 310—HOMEWORK WORKSHEET 06
Matrices, Linear Independence and Gram-Schmidt

1. Write up a **neat** assignment on a **new sheet** of paper. (Do not cram your answers between the lines).
2. **Number** your problems so that it is easy to see what work matches the assigned problems.
3. Be verbose. Remember that you do not understand a concept if you do not know an **examples**.

Gram-Schmidt algorithm for 3 vectors.

Input: $\vec{a}_1, \vec{a}_2, \vec{a}_3$.

Output: either get $\tilde{q}_i = \vec{0}$ for some i , or get an orthonormal list of vectors $\vec{q}_1, \vec{q}_2, \vec{q}_3$.

1. Let $\vec{q}_1 = \frac{1}{\|\vec{a}_1\|} \vec{a}_1$.
2. Let $\tilde{q}_2 = \vec{a}_2 - (\vec{q}_1^T \vec{a}_2) \vec{q}_1$.
3. Let $\vec{q}_2 = \frac{1}{\|\tilde{q}_2\|} \tilde{q}_2$.
4. Let $\tilde{q}_3 = \vec{a}_3 - (\vec{q}_1^T \vec{a}_3) \vec{q}_1 - (\vec{q}_2^T \vec{a}_3) \vec{q}_2$.
5. Let $\vec{q}_3 = \frac{1}{\|\tilde{q}_3\|} \tilde{q}_3$.

Problems

1. Let the columns of matrix A be $\vec{a}_1, \dots, \vec{a}_6$.

$$A = \begin{bmatrix} 0 & 1 & 2 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Find $\vec{a}_1, \dots, \vec{a}_6$.

2. Why can you instantly say that the columns are linearly dependent?
3. What is the largest possible number of linearly independent columns? (Cite the appropriate result from class. Explain.).
4. Apply Gram-Schmidt to $\vec{a}_1, \vec{a}_2, \vec{a}_3$. Show that these vectors are either linearly independent *or* produce an orthonormal list of vectors $\vec{q}_1, \vec{q}_2, \vec{q}_3$.

(over)

5. Take the non-zero $\vec{q}$ vectors from the last problem and add $\vec{a}_4$. Re-apply Gram-Schmidt. Either $\tilde{q}_4 = \vec{0}$ or you will get a larger independent set of vectors.
6. Take the non-zero $\vec{q}$ vectors from the last problem and add $\vec{a}_5$. Re-apply Gram-Schmidt. Either $\tilde{q}_5 = \vec{0}$ or you will get a larger independent set of vectors.
7. Take the non-zero $\vec{q}$ vectors from the last problem and add $\vec{a}_6$. Re-apply Gram-Schmidt. Either $\tilde{q}_6 = \vec{0}$ or you will get a larger independent set of vectors.
8. Check that the final list of non-zero vectors produced in these steps is orthonormal.
9. What are the indexes of the vectors you “kept”? These correspond to a maximal set of linearly independent columns of A . Write out this linearly independent set of columns of A .